

Regularized Cross-Gram Tradeoffs

Singular-Value Clipping Cannot Beat a Residual Larger Than the Cross Angle

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Abstract

The RH-185 bi-Krylov candidate has small directed residuals only in windows where the right and left temporal spaces are nearly transverse. RH-186 shows that the elementary oblique conditioning gate fails. A natural repair is to clip the small singular values of the cross Gram and accept a controlled loss of biorthogonality. This paper derives the exact tradeoff.

Let $\gamma = \sigma_{\min}(Q_L^* Q_R)$, let ϵ be the larger directed residual, and clip at $\tau > 0$. The effective singular value is $d = \max(\gamma, \tau)$. The regularized duality ratio is γ/d , the duality defect is $1 - \gamma/d$, and the regularized coordinate norm is $1/d$. Their sum obeys the exact identity

$$\left(1 - \frac{\gamma}{d}\right) + \frac{\epsilon}{d} = 1 + \frac{\epsilon - \gamma}{d} \quad (\tau \geq \gamma).$$

Consequently a strict combined contraction exists in this clipping family if and only if $\epsilon < \gamma$. If the residual is larger than the cross angle, clipping can trade conditioning for duality but cannot push the total budget below one.

We sweep 13 clipping thresholds over all 126 RH-185 windows, producing 1,638 records. The smallest residual-to-cross-angle ratio is 10.253; no window has a strict regularized contraction, and the best finite grid gate is 1.0090. This is a sharp branch-specific negative result, not a theorem excluding non-spectral regularization or contour cancellation.

1 Motivation

For balanced frames V, W with cross-angle cosine γ , the natural condition number is γ^{-1} [1, 2]. The twelve RH-185 windows passing the raw residual gate have residual levels much larger than their smallest cross singular values. One might hope to replace the exact dual frame by a bounded approximate dual.

Singular-value clipping is the cleanest target-independent test. It acts on the cross Gram itself, not on fitted eigenvectors. If even this family cannot meet a combined duality/conditioning gate, the next regularization must use additional structure rather than merely truncating small cross angles.

2 Clipped frames

Write

$$H = Q_L^* Q_R = U \operatorname{diag}(\sigma_1, \dots, \sigma_r) V^*, \quad \sigma_r = \gamma > 0. \quad (1)$$

For a clipping level $\tau > 0$, set

$$d_j = \max(\sigma_j, \tau). \quad (2)$$

Use the same singular vectors but replace $\sigma_j^{-1/2}$ by $d_j^{-1/2}$:

$$V_\tau = Q_R V \operatorname{diag}(d_j^{-1/2}), \quad W_\tau = Q_L U \operatorname{diag}(d_j^{-1/2}). \quad (3)$$

Then

$$W_\tau^* V_\tau = \operatorname{diag}\left(\frac{\sigma_j}{d_j}\right). \quad (4)$$

The smallest singular direction controls the worst duality defect and the largest clipped inverse controls the coordinate norm. The following theorem is the exact scalar statement needed for the audit.

3 The clipping tradeoff theorem

Theorem 3.1 (Exact regularized budget). *Let $\gamma = \sigma_r$ and let $\epsilon \geq 0$ be a residual level. For the clipped pair (3), define*

$$\delta_\tau = 1 - \frac{\gamma}{\max(\gamma, \tau)}, \quad (5)$$

$$\chi_\tau = \frac{1}{\max(\gamma, \tau)}, \quad (6)$$

$$B_\tau = \delta_\tau + \epsilon \chi_\tau. \quad (7)$$

If $\tau \geq \gamma$, then

$$B_\tau = 1 + \frac{\epsilon - \gamma}{\tau}. \quad (8)$$

There exists $\tau > 0$ with $B_\tau < 1$ if and only if $\epsilon < \gamma$.

Proof. For $\tau \leq \gamma$, the pair is unaltered in the worst direction and $B_\tau = \epsilon/\gamma$. For $\tau \geq \gamma$, substitute $\delta_\tau = 1 - \gamma/\tau$ and $\chi_\tau = 1/\tau$ to obtain (8). If $\epsilon < \gamma$, the unaltered choice gives $B_\tau = \epsilon/\gamma < 1$. If $\epsilon = \gamma$, the value is one for $\tau \leq \gamma$ and approaches one from above for $\tau > \gamma$. If $\epsilon > \gamma$, the unaltered value exceeds one and (8) is strictly larger than one. Thus the stated equivalence holds. $\square$

Corollary 3.2 (Cross-angle criterion). *Within singular-value clipping, regularization closes the combined duality/conditioning gap exactly when*

$$\epsilon < \sigma_{\min}(Q_L^* Q_R). \quad (9)$$

The result is a Pareto identity, not a numerical heuristic. It says that clipping cannot create transversality that the residual has already lost.

The same conclusion persists when the residual information is resolved in the singular directions rather than replaced by one maximum.

Proposition 3.3 (Directionwise clipping criterion). *Let $\epsilon_j \geq 0$ be a residual bound in the j th cross-Gram singular direction and set $d_j = \max(\sigma_j, \tau)$. The directionwise additive budget*

$$B_j(\tau) = 1 - \frac{\sigma_j}{d_j} + \frac{\epsilon_j}{d_j} \quad (10)$$

is strictly below one for every j at some clipping level $\tau > 0$ if and only if

$$\epsilon_j < \sigma_j \quad \text{for every } j. \quad (11)$$

Proof. If (11) holds, choose $\tau \leq \sigma_r$. Then no direction is clipped and $B_j = \epsilon_j/\sigma_j < 1$. Conversely, if $\epsilon_k \geq \sigma_k$ for some k , then for $\tau \leq \sigma_k$ one has $B_k = \epsilon_k/\sigma_k \geq 1$, while for $\tau > \sigma_k$ one has $B_k = 1 + (\epsilon_k - \sigma_k)/\tau \geq 1$. $\square$

Thus the worst-direction theorem is not an artifact of collapsing all singular vectors into one scalar. Direction-dependent clipping can help only when every affected residual direction is already smaller than its available cross angle.

4 Physical sweep

For each RH-185 window we set

$$\epsilon = \max(\epsilon_R, \epsilon_L) \quad (12)$$

and evaluate 13 thresholds $\tau/\gamma \in \{0.25, 0.5, 1, 2, \dots, 1024\}$. This gives 1,638 threshold/window records. The key values are:

quantity	value
windows	126
thresholds per window	13
minimum ϵ/γ	10.2530
median ϵ/γ	902.203
maximum ϵ/γ	1.5855×10^8
minimum finite-grid B_τ	1.00904
strict regularized gates	0

The twelve RH-185 raw gate-passing windows are not exceptions: their residual-to-cross-angle ratios are all larger than one. In particular the best right-channel $L = 4$ window has residual level about 0.0419 and $\gamma \approx 0.00104$, giving a ratio near 40. The best overall ratio, 10.253, still misses the exact criterion by an order of magnitude.

5 What the negative result means

The theorem rules out one very specific repair:

Clip the cross-Gram singular values and pay for the lost duality in a single additive budget.

It does not rule out:

- a contour resolvent that exploits nonnormal cancellation;
- a directed Schur product using separate left and right couplings;
- a different physical dual seed or delayed start;
- a regularizer based on the dynamics rather than cross-Gram singular values alone.

The next paper therefore studies the directed couplings before inserting a complement resolvent.

6 A useful diagnostic identity

For $\tau \geq \gamma$, equation (8) can be rearranged as

$$B_\tau - 1 = \frac{\epsilon - \gamma}{\tau}. \quad (13)$$

The sign of every regularized gate is already decided by the unregularized residual-to-angle comparison. A sweep is still useful for quantifying the distance to one and for detecting coding errors, but it cannot change that sign.

7 Pareto geometry and limiting regimes

For $\tau \geq \gamma$, the two contributions move in opposite directions:

$$\delta_\tau = 1 - \frac{\gamma}{\tau} \nearrow 1, \quad \epsilon\chi_\tau = \frac{\epsilon}{\tau} \searrow 0. \quad (14)$$

If $\epsilon > \gamma$, their sum decreases monotonically toward one from above. This explains why a sufficiently large clipping threshold can make a finite sweep appear numerically close to success without ever producing a strict certificate. If $\epsilon < \gamma$, the unregularized value is already below one and additional clipping increases the sum toward one. At $\epsilon = \gamma$, the sum is exactly one for every threshold.

Accordingly there is no hidden optimum to locate numerically. The only substantive datum is the sign of $\epsilon - \gamma$; the threshold controls how the same failure or success is displayed. In the physical sweep the best finite-grid value 1.00904 is therefore consistent with, rather than contrary to, the exact no-contraction theorem.

8 Reproducibility and branch scope

The archive stores the original cross angle and residual level together with all thirteen thresholds, effective singular values, duality defects, regularized norm products, and combined gates. Recomputing the sign from the original pair (γ, ϵ) independently of the sweep is an internal consistency check supplied by Theorem 3.1.

The negative result is intentionally modular. It can be reused whenever a future packet proposal suggests the same clipping repair: first compare its residual with its cross angle. If the comparison fails, no threshold search in this additive model is needed. If it passes, the unregularized balanced pair already supplies the smallest gate in the family.

9 Boundary

This paper proves the exact clipping tradeoff and verifies it over 1,638 finite physical records. It does not prove that all possible regularizers fail, nor does it prove an all-level cross-angle obstruction. The directional Schur product, exact Feshbach factorization, and contour-specific complement resolvent remain open. R, Gate A, Hilbert–Polya, and RH remain untouched.

References

- [1] Liang Wang. Balanced biorthogonal temporal realization, 2026. Prime Dynamics Theory Program, RH-184.
- [2] Liang Wang. Oblique conditioning in the temporal riesz budget, 2026. Prime Dynamics Theory Program, RH-186.